

Understanding Positive and Negative Transfer in Class-Conditional Diffusion Models

Abstract

Class-conditional diffusion models are often trained on multiple classes and then sampled from a target class by conditioning on the desired label. Empirically, such models may outperform a target-class-only unconditional diffusion model because auxiliary classes can provide additional information about shared visual structures. However, auxiliary classes may also hurt target-class generation when their structures are mismatched with the target class. This project proposes a theoretical framework for understanding when class-conditional diffusion improves target-class generation and when it leads to negative transfer. The central idea is to decompose the class-conditional score into a shared component and a class-specific component. We propose a finite-sample risk decomposition for target-class score estimation that separates shared-structure estimation error, target-specific estimation error, and transfer bias. We further study a Gaussian class-conditional diffusion model as a tractable setting where the benefit of auxiliary classes can be analyzed explicitly. In addition to the main setting, we will also investigate a stronger comparison: whether auxiliary classes can replace the same number of true target-class samples. The goal is not to propose a new diffusion architecture, but to provide a statistical explanation of positive and negative transfer in conditional diffusion models.

1. Introduction

Diffusion models are often trained in a conditional form. For example, a class-conditional diffusion model may be trained jointly on images from dogs, cats, birds, and other classes, and then sampled only from the dog class. A natural question is whether the conditional model trained with multiple classes should outperform an unconditional model trained only on dog images.

There is no general reason why the answer must always be yes. Auxiliary classes may improve generation by helping the model learn shared structures such as edges, textures, shapes, or low-level image statistics. However, if the auxiliary classes are too different from the target class, the shared representation learned by the model may become biased toward irrelevant structures, leading to negative transfer. Thus, the main question is not whether conditional diffusion is always better, but:

When do auxiliary classes improve target-class generation in class-conditional diffusion models, and when do they hurt?

This proposal develops a theoretical framework for this question. The proposed work focuses on score-estimation risk. Since diffusion models are trained by learning denoising functions or score functions across noise levels, score-estimation error provides a natural statistical object for studying finite-sample behavior.

2. Background

2.1 Diffusion models

Let $x_0 \sim p_{\text{data}}$ denote a data sample. A diffusion model defines a forward noising process

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

where $\{\beta_t\}_{t=1}^T$ is a variance schedule. Equivalently, for

$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

we have the closed-form noising distribution

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

or

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

The reverse process is learned as

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

A commonly used parameterization trains a neural network ϵ_θ to predict the noise ϵ :

$$\mathcal{L}_{\text{DDPM}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t) \right\|_2^2 \right]$$

This objective can be viewed as a form of denoising score matching across multiple noise levels.

2.2 Score-based

For a distribution p_t of noisy data x_t , the score function is

$$s_t^*(x) = \nabla_x \log p_t(x)$$

In score-based generative modeling, one estimates $s_t^*(x)$ for multiple noise levels and uses the estimated score to gradually transform noise into data. The denoising network in diffusion models is closely related to the score by

$$s_t^*(x_t) \approx -\frac{1}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_\theta(x_t, t)$$

Thus, analyzing diffusion training can be reduced to analyzing how well the model estimates the score function at each diffusion time.

2.3 Class-conditional diffusion

Suppose data are drawn from K classes. Let $c \in \{1, \dots, K\}$ denote the class label, and let

$$x_0 \mid c \sim p_c$$

A class-conditional diffusion model estimates

$$s_c^*(x, t) = \nabla_x \log p_{t,c}(x)$$

where $p_{t,c}$ is the distribution of x_t when $x_0 \sim p_c$.

The class-conditional denoising objective can be written as

$$\mathcal{L}_{\text{cond}}(\theta) = \mathbb{E}_{c, x_0, \epsilon, t} \left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|_2^2 \right]$$

where

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$$

The central issue is that the target class may benefit from additional classes through shared score components, but may also suffer from transfer bias when the auxiliary classes are not aligned with the target class.

3. Research Question

Let class 1 be the target class. We compare two approaches:

- (1) **Target-only unconditional diffusion:** train an unconditional diffusion model using only samples from class 1.
- (2) **Multi-class conditional diffusion:** train a class-conditional diffusion model using samples from all classes, then sample only from class 1.

The main research question is:

Under what statistical conditions does the multi-class conditional diffusion model achieve smaller target-class score-estimation risk than the target-only unconditional model?

A more precise version is:

$$R_1(\hat{s}_{\text{cond}}) < R_1(\hat{s}_{\text{uncond}})$$

where:

$$R_1(\hat{s}_{\text{cond}}) = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} \left[\|\hat{s}_{\text{cond}}(X_t, t, 1) - s_1^*(X_t, t)\|_2^2 \right] dt$$

$$R_1(\hat{s}_{\text{uncond}}) = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} \left[\|\hat{s}_{\text{uncond}}(X_t, t) - s_1^*(X_t, t)\|_2^2 \right] dt$$

Here $w(t)$ is a nonnegative weight function over diffusion time.

4. Proposed Theoretical Framework

4.1 Shared and class-specific score decomposition

The key modeling assumption is that the class-conditional score can be decomposed into a shared component and a class-specific component:

$$s_c^*(x, t) = s_{\text{shared}}^*(x, t) + s_{\text{specific},c}^*(x, t)$$

Here s_{shared}^* captures structures common across classes, while $s_{\text{specific},c}^*$ captures features unique to class c .

For example, in image generation, the shared component may represent low-level image statistics such as edges, textures, and local smoothness, while the class-specific component may represent the shape and semantic structure of dogs, cats, cars, or airplanes.

A conditional diffusion model trained on multiple classes can use all samples to estimate the shared component. In contrast, a target-only model must estimate both shared and target-specific structures using only target-class samples.

4.2 Target-class score-estimation risk

For the target class $c = 1$, define the risk of an estimator $\hat{s}$ as

$$R_1(\hat{s}) = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} \left[\|\hat{s}(X_t, t, 1) - s_1^*(X_t, t)\|_2^2 \right] dt$$

Suppose the conditional model estimates the score as

$$\hat{s}_{\text{cond}}(x, t, 1) = \hat{s}_{\text{shared}}(x, t) + \hat{s}_{\text{specific},1}(x, t)$$

Then the error is

$$\hat{s}_{\text{cond}}(x, t, 1) - s_1^*(x, t) = [\hat{s}_{\text{shared}}(x, t) - s_{\text{shared}}^*(x, t)] + [\hat{s}_{\text{specific},1}(x, t) - s_{\text{specific},1}^*(x, t)]$$

This motivates the following decomposition:

$$R_1(\hat{s}_{\text{cond}}) = R_{\text{shared}} + R_{\text{specific}} + R_{\text{interaction}}$$

where

$$R_{\text{shared}} = \int_0^T w(t) \mathbb{E}_{p_{t,1}} \left[\|\hat{s}_{\text{shared}}(X_t, t) - s_{\text{shared}}^*(X_t, t)\|_2^2 \right] dt$$

$$R_{\text{specific}} = \int_0^T w(t) \mathbb{E}_{p_{t,1}} \left[\|\hat{s}_{\text{specific},1}(X_t, t) - s_{\text{specific},1}^*(X_t, t)\|_2^2 \right] dt$$

and

$$R_{\text{interaction}} = 2 \int_0^T w(t) \mathbb{E}_{p_{t,1}} \left[\langle \hat{s}_{\text{shared}}(X_t, t) - s_{\text{shared}}^*(X_t, t), \hat{s}_{\text{specific},1}(X_t, t) - s_{\text{specific},1}^*(X_t, t) \rangle \right] dt$$

4.3 Transfer bias

The previous decomposition assumes that all classes share the same s_{shared}^* . In practice, the shared structure may only be approximately shared across classes. Let $s_{\text{shared},1}^*$ denote the target-class shared component, and let $s_{\text{shared,aux}}^*$ denote the population shared component learned from the pooled multi-class data. We define the transfer bias as

$$b_{\text{aux}}(x, t) = s_{\text{shared,aux}}^*(x, t) - s_{\text{shared},1}^*(x, t).$$

Training on all classes, conditional model may not converge to the target shared component, but rather converge to some kind of auxiliary shared component $s_{\text{shared,aux}}^*$.

So the shared estimation error can be written as

$$\eta_{\text{shared}}(x, t) = \widehat{s}_{\text{shared}}(x, t) - s_{\text{shared,aux}}^*(x, t),$$

and the target-specific estimation error as

$$\eta_{\text{specific}}(x, t) = \widehat{s}_{\text{specific},1}(x, t) - s_{\text{specific},1}^*(x, t).$$

Then the target-class score error decomposes as

$$\begin{aligned} \widehat{s}_{\text{cond}}(x, t, 1) - s_1^*(x, t) &= [\widehat{s}_{\text{shared}}(x, t) - s_{\text{shared},1}^*(x, t)] + [\widehat{s}_{\text{specific},1}(x, t) - s_{\text{specific},1}^*(x, t)] \\ &= \eta_{\text{shared}}(x, t) + \eta_{\text{specific}}(x, t) + b_{\text{aux}}(x, t). \end{aligned}$$

Consequently, the target risk contains the transfer-bias term

$$B_{\text{aux}} = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} [\|b_{\text{aux}}(X_t, t)\|_2^2] dt,$$

$$R_1(\widehat{s}_{\text{cond}}) = R_{\text{shared}}^{\text{est}} + R_{\text{specific}}^{\text{est}} + B_{\text{aux}} + R_{\text{cross}},$$

$$R_{\text{shared}}^{\text{est}} = \int_0^T w(t) \mathbb{E}_{p_{t,1}} [\|\eta_{\text{shared}}(X_t, t)\|_2^2] dt,$$

$$R_{\text{specific}}^{\text{est}} = \int_0^T w(t) \mathbb{E}_{p_{t,1}} [\|\eta_{\text{specific}}(X_t, t)\|_2^2] dt,$$

$$B_{\text{aux}} = \int_0^T w(t) \mathbb{E}_{p_{t,1}} [\|b_{\text{aux}}(X_t, t)\|_2^2] dt,$$

$$\begin{aligned} R_{\text{cross}} &= 2 \int_0^T w(t) \mathbb{E}_{p_{t,1}} [\langle \eta_{\text{shared}}(X_t, t), \eta_{\text{specific}}(X_t, t) \rangle \\ &\quad + \langle \eta_{\text{shared}}(X_t, t), b_{\text{aux}}(X_t, t) \rangle \\ &\quad + \langle \eta_{\text{specific}}(X_t, t), b_{\text{aux}}(X_t, t) \rangle] dt. \end{aligned}$$

5. Main Theoretical Goal

The main theoretical goal of this project is to turn the risk decomposition in Section 4 into a finite-sample comparison between multi-class conditional diffusion and target-only unconditional diffusion. We aim to control the four terms that arise from the target-class risk decomposition:

$$R_1(\widehat{s}_{\text{cond}}) = R_{\text{shared}}^{\text{est}} + R_{\text{specific}}^{\text{est}} + B_{\text{aux}} + R_{\text{cross}}.$$

Here $R_{\text{shared}}^{\text{est}}$ measures the finite-sample estimation error of the shared score component, $R_{\text{specific}}^{\text{est}}$ measures the finite-sample estimation error of the target-specific score component, B_{aux} measures the transfer bias induced by auxiliary-class mismatch, and R_{cross} contains the cross terms among these components.

5.1 Controlling the four terms

Let n_1 denote the number of target-class samples and let

$$N = \sum_{c=1}^K n_c$$

denote the total number of samples used by the multi-class conditional model. The conditional model uses both target and auxiliary samples, while the target-only unconditional model uses only the n_1 target-class samples.

For a training dataset $\mathcal{D}$, the learned score estimator is $\hat{s}_{\mathcal{D}}$. We define the conditional target-class risk as

$$R_1(\hat{s}_{\mathcal{D}}) = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} [\|\hat{s}_{\mathcal{D}}(X_t, t) - s_1^*(X_t, t)\|_2^2] dt.$$

This is the population risk conditional on the training dataset. Since $\hat{s}_{\mathcal{D}}$ is random through $\mathcal{D}$, we study the expected risk

$$\mathbb{E}_{\mathcal{D}} R_1(\hat{s}_{\mathcal{D}})$$

We aim to prove bounds of this form

$$\begin{aligned} \mathbb{E}_{\mathcal{D}} R_{\text{shared}}^{\text{est}} &\leq \mathcal{E}_{\text{shared}}(N), \\ \mathbb{E}_{\mathcal{D}} R_{\text{specific}}^{\text{est}} &\leq \mathcal{E}_{\text{specific}}(n_1), \end{aligned}$$

and

$$|\mathbb{E}_{\mathcal{D}} R_{\text{cross}}| \leq \mathcal{C}_{\text{cross}}(N, n_1, B_{\text{aux}}),$$

where $\mathbb{E}_{\mathcal{D}}$ denotes expectation over the training data. The function $\mathcal{E}_{\text{shared}}(N)$ represents the statistical error of estimating the shared score component using all N samples, while $\mathcal{E}_{\text{specific}}(n_1)$ represents the statistical error of estimating the target-specific component using only the n_1 target samples. The quantity $\mathcal{C}_{\text{cross}}$ controls the cross terms. In some idealized settings, such as orthogonal linear models or Gaussian models with zero-mean errors, this term may vanish.

Under such controls, the conditional target-class risk is expected to satisfy a bound of the form

$$\mathbb{E}_{\mathcal{D}} R_1(\hat{s}_{\text{cond}}) \lesssim \mathcal{E}_{\text{shared}}(N) + \mathcal{E}_{\text{specific}}(n_1) + B_{\text{aux}} + \mathcal{C}_{\text{cross}} + \mathcal{A}_{\text{cond}},$$

where $\mathcal{A}_{\text{cond}}$ denotes approximation or optimization errors not captured by the idealized decomposition.

By contrast, the target-only unconditional model must estimate both shared and target-specific structures using only n_1 target-class samples. Thus, we aim to compare the above bound with a target-only bound of the form

$$\mathbb{E}_{\mathcal{D}} R_1(\hat{s}_{\text{uncond}}) \lesssim \mathcal{E}_{\text{shared}}(n_1) + \mathcal{E}_{\text{specific}}(n_1) + \mathcal{C}_{\text{uncond}} + \mathcal{A}_{\text{uncond}}.$$

These bounds should not be interpreted as a direct proof that the conditional risk is smaller than the unconditional risk, since comparing two upper bounds alone is not sufficient to compare the true risks. In concrete models, we aim to derive either exact risk expressions, or upper and lower bounds.

5.2 Positive-transfer goal

The first goal is to identify conditions under which

$$R_1(\hat{s}_{\text{cond}}) < R_1(\hat{s}_{\text{uncond}}).$$

In the proposed framework, positive transfer occurs when the auxiliary classes are sufficiently aligned with the target class so that B_{aux} is small, and when the additional auxiliary data substantially reduces the shared-component estimation error.

5.3 Negative-transfer goal

The second goal is to find the condition when auxiliary classes hurt target-class generation. Negative transfer is expected when the auxiliary classes are poorly aligned with the target class, causing the shared component learned by the conditional model to be biased away from the target-class shared component.

In terms of the decomposition, negative transfer occurs when B_{aux} and the associated cross terms dominate the statistical gain from reducing $R_{\text{shared}}^{\text{est}}$. This explains why training on more classes is sometimes not better for a fixed target class. Even though the conditional model has access to more total data, the non-target data can pull the learned score field toward directions that are inappropriate for the target class.

5.4 Potential Corollary: can auxiliary classes replace target-class data?

The main comparison in this proposal is:

conditional model trained on n_1 target samples and $N - n_1$ auxiliary samples

vs

target-only unconditional model trained on n_1 target samples.

A stronger comparison can also be studied as a secondary question. Instead of comparing against a target-only model trained with only n_1 target samples, we may compare against an target-only model trained with N target-class samples:

Can auxiliary classes replace the same number of true target-class samples?

This can be viewed as a simple modification of the main target-only unconditional baseline. In the main comparison, the unconditional baseline is trained with only n_1 target-class samples. For the stronger comparison, we replace this sample size by N , as if the unconditional model had access to N true target-class samples:

$$R_{\text{target}}(n_1) \longrightarrow R_{\text{target}}(N).$$

This asks whether the $N - n_1$ auxiliary samples used by the conditional model can serve as a substitute for the same number of additional target-class samples. This is a stronger benchmark, and it helps clarify the limitation of auxiliary classes: auxiliary data may help in low-target-data regimes, but they may not fully replace true target-class data when target-specific information is important.

5.5 Role of concrete models

The abstract formulation above leaves the terms $\mathcal{E}_{\text{shared}}$, $\mathcal{E}_{\text{specific}}$, B_{aux} , and $\mathcal{C}_{\text{cross}}$ unspecified. The exact form of these terms depends on the choice of model class, distributional assumptions, etc.

A natural next step is to specialize the framework in simple models. For example, one can first study a Gaussian class-conditional model in which different classes share a common covariance structure but have different means. In this setting, the shared component can be associated with the common covariance, while the class-specific component can be associated with the target-class mean. This model provides a clean test case for analyzing how auxiliary classes reduce the estimation error of shared structure.

6. Related Work: Conditional Generative Modeling on Multiple Data Sources

A closely related theoretical work is [1], which studies conditional generative modeling on multiple data sources. In their setting, each condition $Y = k$ represents a distinct data source, and the true conditional distribution is assumed to have the form

$$p_{X|k}^*(x|k) = p_{\phi_k^*, \psi^*}(x|k),$$

where ϕ_k^* is a source-specific parameter and ψ^* is a parameter shared across all sources. They compare two training strategies. The first is multi-source training, which trains one conditional generative model on all sources jointly and shares ψ across sources. The second is single-source training, which trains a separate model for each source, with its own source-specific and shared-like parameters.

Their main theoretical contribution is an average total-variation error bound for conditional maximum likelihood estimation. In particular, they show that when source distributions share parametric similarities and the model class is realizable, multi-source training can have a sharper average distribution estimation bound than single-source training. The improvement arises because sharing parameters reduces the complexity of the conditional distribution space. They instantiate this theory in conditional Gaussian models, autoregressive models, and energy-based models, and they validate the theory using both Gaussian simulations and class-conditional diffusion experiments.

This work is highly relevant to our proposal because both settings involve multiple classes and both rely on the idea that different sources may share common structure. However, the focus of our project is different in several important ways. First, [1] studies average performance over all sources, whereas we focus on a fixed target class $c = 1$. Our main object is the target-class score-estimation risk

$$R_1(\hat{s}) = \int_0^T w(t) \mathbb{E}_{X_t \sim p_{t,1}} \left[\|\hat{s}(X_t, t) - s_1^*(X_t, t)\|_2^2 \right] dt,$$

rather than an average total-variation distance over all sources. Second, their comparison is between one multi-source conditional model and K separate single-source models, while our comparison is between a multi-class conditional diffusion model sampled at the target label and a target-only unconditional diffusion model trained only on target-class data. Third, their theory mainly establishes when multi-source training is beneficial under source similarity, whereas our goal is to characterize both positive transfer and negative transfer for a fixed target class.

Therefore, our intended contribution is not the general idea of a shared and source-specific decomposition, which already appears in multi-source conditional generative modeling. Instead, our contribution is to formulate a diffusion-specific target-class transfer problem at the score level. We aim to decompose the target-class score risk into shared estimation error, target-specific estimation error, auxiliary-class transfer bias, and cross terms. This allows us to ask a different question:

For a fixed target class, which auxiliary classes help target-class diffusion generation, and which auxiliary classes hurt it?

This target-class positive/negative transfer perspective is not directly addressed by existing average multi-source distribution-estimation theory.

7. Expected Contributions

This project aims to make the following contributions:

- (1) Develop a theoretical framework for studying target-class generation in class-conditional diffusion models.
- (2) Formulate a score-level shared-specific decomposition for fixed-target auxiliary-class transfer in class-conditional diffusion models.
- (3) Derive a finite-sample risk decomposition separating shared-structure estimation error, target-specific estimation error, and transfer bias.
- (4) Prove positive-transfer and negative-transfer conditions in terms of target sample size, auxiliary sample size, effective dimension, and class mismatch.
- (5) Provide a rigorous Gaussian theory showing how auxiliary classes improve estimation of shared covariance structure.

The main novelty is theoretical rather than architectural. Instead of proposing another diffusion-model variant, this project seeks to explain an important empirical phenomenon: why class-conditional diffusion sometimes improves target-class generation, and why it can also fail when auxiliary classes are poorly matched.

7. Broader Significance

In many applications, target-class data may be scarce, while auxiliary data from related classes are abundant. A principled theory can guide when to train a conditional model using auxiliary data and when to avoid harmful transfer.

References

- [1] Rongzhen Wang, Yan Zhang, Chenyu Zheng, Chongxuan Li, and Guoqiang Wu. A Theory for Conditional Generative Modeling on Multiple Data Sources. In *International Conference on Machine Learning*, 2025.